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Power distribution system fault cause analysis by using association rule mining

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ABSTRACT

In recent years, with the increasing requirements on power distribution utilities to ensure system reliability and to improve customers and regulators satisfaction, utilities seek to find practical solutions that enable them to restrict specific faults or to better manage their responses to unavoidable power outages. For achieving either, it is crucial to acquire a profound understanding of different faults by exploring their underlying causes and identifying key variables related to those causes. Currently, statistical models as well as advanced data analytics techniques are common tools to gain such understanding. Although basic statistical analysis provides a general knowledge of the primary causes of faults; nevertheless, it falls short of describing nuanced conditions that lead to a fault. On the other hand, applying sophisticated algorithms can produce deeper insight into the main causes; however, it would be computationally burdensome and might require a tremendous amount of running time. In order to overcome these problems, this paper proposes a novel approach for fault cause analysis by using association rule mining. The primary goals are to characterize faults according to their underlying causes and to identify important variables that strongly impact fault frequency. This paper proposes a step-by-step procedure, which deals with data preparation, practical issues associated with fault data sets, and implementation of association rule mining. The procedure is followed by a comprehensive case study to demonstrate how the proposed approach can be used to mine for causal structures and identify frequent patterns for vegetation, animal, equipment failure, public accident, and lightning-related faults.

1. Introduction

1.1. Motivation

Faults in power distribution systems can seriously endanger the system operation in different ways. As a matter of fact, distribution faults negatively influence the system reliability since they are responsible for a considerable number of major interruptions that customers experience [1]. Furthermore, faults exert damaging impacts on system safety and security and result in heavy costs for distribution utilities [2]. Therefore, utilities either seek to find practical solutions aimed at preventing specific faults or attempt to take effective measures to properly and quickly restore the system after faults occur. For attaining either, it is essential to acquire a deeper understanding of the primary causes of faults and to identify significant variables related to those causes.

Power distribution utilities are usually interested in preventing avoidable faults. One main course of action to fulfill this objective is to introduce necessary modifications to system fault management based on the knowledge acquired from faults that have occurred in the past. For instance, if it turns out that a distribution system had experienced most of its vegetation-related faults during the months of June and July, then the utility will recognize the need to carry out essential preventive maintenance for those two months. In addition to modifying fault management practices, such knowledge is highly beneficial for improving the design of existing distribution systems to reduce the number of future faults [3]. In order to gain this knowledge, it is required to carry out an in-depth root cause analysis for different faults.

Nevertheless, most often, distribution faults have proved to be unavoidable. For example, even with the implementation of different preventive measures such as installing animal guards or designing large clearances between phase and ground wires, substantial numbers of animal-related faults occur in the system. In these cases, distribution utilities attempt to take an appropriate response to the fault either by predicting it or identifying the causes immediately after the fault. The task of predicting or identify-

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ing faults has been considered extremely challenging due to the random nature of faults and the numerous contributing factors. However, over the recent years, with the explosion in data gathering within the smart grid framework, applications of advanced data analytics techniques combined with the traditional rigorous mathematical modeling have facilitated the tasks of fault prediction and identification [4]. In fact, these applications have been identified as viable technologies with the capability to provide necessary actionable information support for utilities to predict or identify different faults with satisfactory accuracy [5–7]. For the purpose of developing these models, however, it is crucial to identify and utilize the factors that are related to each fault cause. For example, the authors in [6] consider six features of: weather condition, season, time of day, faulty phases, protective device activated, and the location where the fault happened as the most influential factors for vegetation and animal-related faults and use them as inputs to build a power distribution fault cause identifier.

Consequently, characterizing faults according to their underlying causes and identifying significant variables that strongly impact the fault frequency are extremely valuable as they allow utilities to find solutions to restrict specific causes and to give an appropriate response to unavoidable faults.

1.2. Literature review

By this time, several studies have been conducted to analyze the characteristics of various faults, which have been caused primarily by animals or vegetation-related issues. These studies mainly take advantage of statistical techniques and data analytics algorithms.

For instance, the authors in [3] propose a statistical data mining approach to perform a root cause analysis of animal-related faults. In their work, the impact of six factors of weather condition, season, the day of the week, time of day, faulty phases, and protection device activated is considered to be significant on the frequency of such faults. In another attempt, by utilizing four statistical measures of actual, normalized, relative, and likelihood, the authors in [8] investigate tree-related faults with respect to six factors: weather condition, season, time of day, the number of faulty phases, location and the clearing device. Furthermore, the same authors in [9] use logistic regression to explore the influence of the aforementioned factors on tree faults by evaluating significance levels resulting from regression. In their study, the weather condition is observed to be the most influential factor. In [10], the authors review two statistical methods, namely hypothesis test, and stepwise regression and introduce two new methods to select proper features for identifying root causes of different faults. They employ these methods to examine six factors explained in [6].

1.3. Contributions

Although basic statistical analysis provides a general understanding of the primary causes of faults, it falls short of describing nuanced conditions that lead to a fault. In fact, gaining a deeper understanding of the contributing factors for each fault type by using conventional statistical methods could become extremely time consuming as it requires performing various analyses. On the other hand, applying sophisticated methods such as stepwise and logistic regression or artificial neural networks can produce deeper insight into the underlying causes; however, it would be computationally burdensome and might require a tremendous amount of running time [10]. The reason for the extra computation time requirement is that a substantial number of inputs, which are based on the many attributes that are involved in the study, have to be generated and fed into these methods.

In order to overcome the aforementioned problems, this paper proposes an approach for fault cause analysis based on association rule mining. Association rules, introduced by [11], belong to a category of uncomplicated but remarkably powerful regularities in binary data. They initially were employed to mine large collections of basket data type transactions for association rules between sets of items. However, due to their proven capabilities for finding interesting associations, or correlations among data items, they have received increasing attention in different fields for practical applications. The main advantage of association rule mining over the aforementioned methods is that it can easily analyze the co-occurrence of different attributes to find any association or correlation. Also, since association rule algorithms were originally developed to be applied to extremely large transaction data sets, they are very effective with regard to the computation time requirement. In fact, as will be demonstrated in the following sections of this paper, the association rule mining is capable of extracting comprehensive patterns from fault data sets within a short amount of time. To implement it, however, in-depth data preparation is required. Moreover, there are several practical issues associated with fault data sets that ought to be addressed. Therefore, this paper provides a step-by-step procedure that fully deals with necessary data preparation, practical issues associated with fault data sets, and implementation of association rule mining. Furthermore, this procedure is applied to investigate a real-world fault data set. As a result of the case study, causal structures and frequent patterns for vegetation, animal, equipment failure, public accident, and lightning-related faults, which have drawn less attention previously in the literature are explored.

1.4. Paper organization

This paper is organized as follows. Association rule mining is discussed in Section 2. In Section 3, the problem of data insufficiency is explained, and a practical solution to deal with it is provided. The proposed methodology is elaborated in Section 4. Section 5 provides a case study to illustrate the implementation of the proposed methodology. Finally, Section 6 draws conclusions on the effectiveness of association rule mining in fault cause analysis.

2. Association rule mining

As mentioned earlier, association rules were initially introduced to mine large collections of basket data type transactions to investigate how items or objects are related to each other. However, nowadays, because of their effectiveness, they are widely employed in different domains to mine for causal structures and to identify frequent patterns in various data sets.

An association rule is a causality, where a rule is defined as an implication of the form $X \Rightarrow Y$, with two conditions of $X, Y \subseteq I$ and $X \cap Y = \emptyset$, where I is a set of n binary attributes called items [11]. The itemsets X and Y are called antecedent (left-hand-side or LHS) and consequent (right-hand-side or RHS) of the rule, respectively.

Given the set of transactions, numerous rules can be generated. However, the rules that are of actual interest to data analysts, which provide useful information, have to fulfill certain constraints. The major constraints related to the “support” and “confidence” of a rule [11]. Support determines how often a rule is applicable to a given data set, while confidence determines how frequently items in Y appear in transactions that contain X. Support and confidence can be mathematically expressed as (1) and (2), respectively.

$$Support(X \Rightarrow Y) = \frac{\sigma(X, Y)}{N} \quad (1)$$

$$Confidence(X \Rightarrow Y) = \frac{\sigma(X, Y)}{\sigma(X)} \quad (2)$$

where, σ is summation notation, and N represents the total number of all transactions.

In this formulation, the association rule problem is usually decomposed into two main problems as follows [11].

- Generating frequent itemsets: The first problem is to discover the itemsets whose occurrences surpass a minimum support. These itemsets are called frequent or large itemsets;
- Generating rules: The second problem is to generate association rules from those large itemsets with the constraint of minimal confidence.

While handling the second problem is straightforward, addressing the first problem can be challenging. Actually, finding all frequent itemsets in a database is difficult since it requires searching over all possible itemset combinations [12]. Therefore, different techniques have been proposed to effectively deal with this issue. One of the most well-established techniques with this regard is the Apriori algorithm. Apriori is the first association rule mining algorithm that utilizes the support-based pruning for the purpose of controlling the exponential growth of candidate itemsets systematically [12]. This algorithm is employed in this paper to find frequent itemsets and to generate rules based on confidence.

In order to rank the resultant rules from the Apriori algorithm, there are different measures. As mentioned, support and confidence are the primary metrics to evaluate the quality of the rules generated by the algorithm as they indicate the statistical significance and the strength of a rule, respectively. In addition, there are other metrics, one of which, is "lift" value of a rule. The lift value can be stated as (3)

$$Lift(X \rightarrow Y) = \frac{Support(X \Rightarrow Y)}{Support(Y) \times Support(X)} \quad (3)$$

Lift value is an indicator of the strength of a rule over the random co-occurrence of the antecedent and the consequent [13]. A lift ratio larger than 1.0 implies that the relationship between the antecedent and the consequent is more significant than when the two sets are independent. The larger the lift ratio, the more significant the rule.

3. Generating synthetic data

Over recent years, with the deployment of an enormous number of intelligent electronic devices and various sensors, rich data for studying different faults have become available [14]. However, having access to sufficient amounts of data for pattern recognition is not always possible. Furthermore, a significant challenge with real-world fault data is the lack of adequate information relating to specific types of faults as their occurrence is not frequent. For example, only 3% of faults occurred in and around Charlotte, North Carolina region, between the years of 2009 and 2014 were due to lightning. Hence, if one wants to find patterns associated with lightning faults, one might face the insufficient data problem. In fact, such inadequacy in data can pose serious problems to the efficient operation of association rule mining algorithms as it leads the algorithm to fail to generate requisite large itemsets.

In order to address the aforementioned issue, one practical solution is to generate synthetic data based on the available real data and add them to the dataset [15]. However, this task should be carried out such that hidden complex patterns within the data set are preserved as much as possible. In order to comply with this requirement, synthetic data can be generated based on the Synthetic Minority Over-sampling Technique (SMOTE). This method, introduced in [15], is of particular interest to analysts due to its simplicity and effectiveness. SMOTE has been tested on different datasets, with varying degrees of imbalance and varying amounts of data in the training set, successfully providing a diverse testbed [16]. For the purpose of creating new artificial samples, this method uses the k-nearest neighbor algorithm and bootstrapping [17]. The process for generating artificial data based on this method is illustrated in Fig. 1. As seen in the figure, the procedure includes four steps as follows [15].

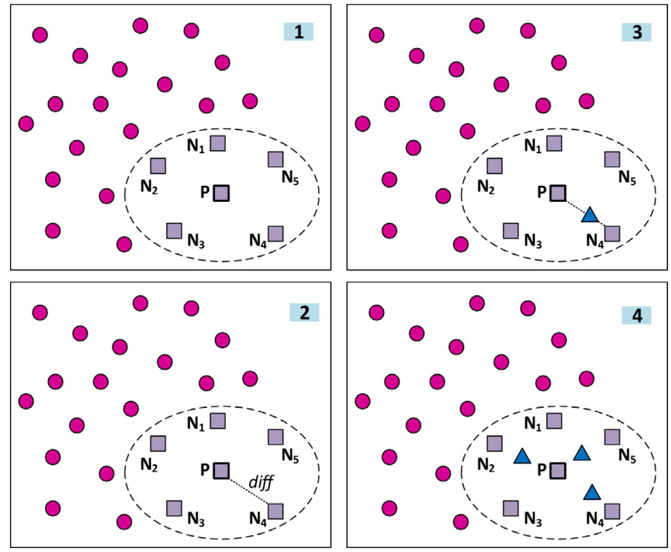

Fig. 1. SMOTE algorithm.

- Step 1: For each minority sample P, its k neighbors (shown with squares) are found (5 neighbors in this example), and an example out of k neighbors (N_4) is randomly chosen;
- Step 2: The difference between the sample under consideration and its selected neighbor is taken;
- Step 3: The difference is multiplied by a random number between 0 and 1 to generate synthetic data (shown with triangles);
- Step 4: The generated synthetic data are added to the dataset.

Synthetic data that are generated by this method mimic the original observed data and preserve the relationships between variables; consequently, they are considered as one feasible solution to the problem of data insufficiency.

4. Proposed approach

The structure of the proposed approach is illustrated in the flow chart provided in Fig. 2. In what follows, each block of the flow chart is described in detail.

(1) *Data Collection and Preparation*: An essential step in performing fault analysis for pattern recognition is to collect and prepare a considerable amount of fault data. In a typical fault data, one column is assigned to the fault cause (i.e. vegetation, animal, etc.) and the rest of the columns are specified for the features. Such data can be gathered from various sources such as a utility company's electronic devices, sensors, weather stations, etc. However, as mentioned earlier, occasionally having access to a sufficient amount of data is not possible; therefore, the lack of data ought to be artificially compensated.In order to effectively gather and prepare data, this task is split into different categories as follows.

1.1 *Obtaining data for various features*: The most vital part of this task is to obtain data for as many features as possible. In order to analyze a specific fault, a large amount of information pertaining to that fault ought to be available. In fact, without having necessary features that make significant contribution to a specific fault,

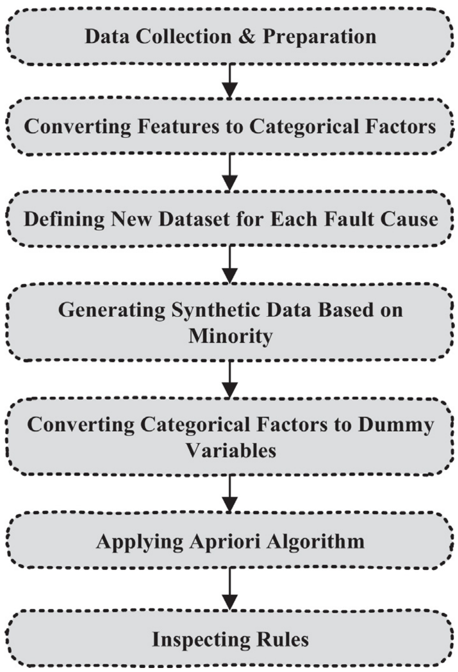

Fig. 2. Technical flow chart of the proposed approach.

understanding the characteristics of that fault would be practically impossible.

1.2 *Data cleansing*: Real-world fault data sets usually contain a significant amount of input errors, duplicate data, outliers, extreme and unexpected values. These errors can perturb the operation of data mining algorithms. Therefore, the raw data should be explored to identify and remove these errors.

1.3 *Dealing with missing values*: Fault datasets typically beset with a substantial number of missing values. The missing data, which can be because of errors in measurements and sensor malfunction, pose serious problems for data analysis. As a result, it is necessary to deal with them. One simplistic approach to address this issue is to remove the rows and columns that contain missing values; nevertheless, this leads to losing valuable information. Therefore, alternative approaches that deal with them properly must be employed.

(2) *Converting continuous features to categorical factors*: Fault data sets are often described with various features that have different formats. Some of them can be continuous, while the others are categorical. However, in order to develop the proposed methodology, it is necessary to convert all continues features to categorical ones. To fulfill this task, those features can be categorized into different groups and then represented by group number. For instance, vegetation-related faults that occurred in Charlotte, North Carolina region, between the years of 2009 and 2014 were associated with different ranges of wind speed. This range continuously varies from 1 m/s to 19 m/s. To make the wind speed feature categorical, it can be categorized into 7 groups as [1,3.5), [3.5,6), [6,8.5), [8.5,11), [11,13.5), [13.5,16), and [16,19] and then represented by categorical factors of 1–7, respectively. The selection of the number of categories depends on the desired resolution.

(3) *Defining new dataset for each fault cause*: The next step is to create a new dataset for each fault cause. In order to perform this, first, the main fault data set is replicated by the number of all fault causes or by the number of fault causes that are the targets for the study. For example, if the purpose is to find patterns for vegetation and animal-related faults, then original data set will be replicated two times. The new data sets are called sub-dataset, which their title is the fault cause. Afterward, the column that shows the fault cause in each sub-dataset is transformed such that it only contains binary values of 1 and 0. The value would be 1 if the fault cause is the same as the title of the sub-dataset; otherwise, it would be 0. An example of this transformation for vegetation-related faults is illustrated in Fig. 3. It is worth mentioning that the features for all sub-datasets are the same. Moreover, all sub-datasets have the same number of samples. The only difference is in the column that shows the fault cause.

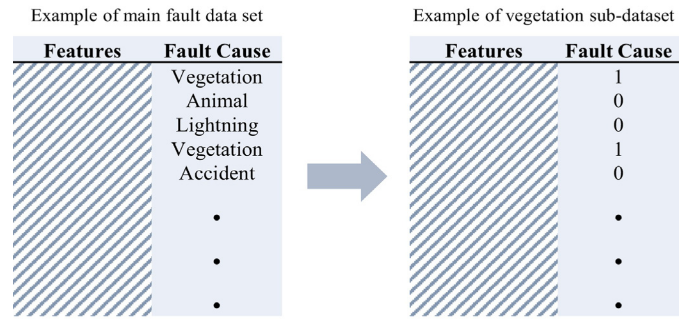

Fig. 3. Illustration of defining sub-dataset for vegetation-related faults.

(4) *Generating synthetic data for each sub-dataset*: In order to address the problem of data insufficiency, synthetic data are to be generated for each sub-dataset. As illustrated in Fig. 1, artificial data are generated based on the minority group. The minority group in each sub-dataset includes those samples in which their fault cause is represented by 1. Therefore, by applying the SMOTE algorithm, the number of these samples is increased such that finally each sub-dataset contain almost the same number of samples that are represented by 0 and 1.

(5) *Converting categorical factors to dummy variables*: Next, for all sub-datasets, all features are to be converted into dummy variables. In order to conduct this, the name of the feature is added to the category number to create a new string, then the former value is replaced with this string. For instance, as mentioned, the wind speed can be converted to categorical factors represented by 1–7. If for a specific fault sample the wind speed is represented by 5, it should be converted into WIND5. Likewise, the same procedure ought to be followed for the column that demonstrates the fault cause. For example, for vegetation sub-dataset, if the value for fault cause is 1, it should be converted to VEGETATION1. Taking this step is necessary for interpreting the rules generated by association mining.

(6) *Mining association rules for each sub-dataset*: By following the procedure explained thus far, different sub-datasets are generated that resemble basket data type transactions; therefore, they are qualified to be explored by association rule mining. In order to perform association rule analysis, the Apriori algorithm is utilized for each sub-dataset. For this reason, different constraints are to be first determined as follows.

6.1 *Minimum support*: In order to find frequent itemsets, the minimum support value should be defined. In general, selection of this value depends on the dataset characteristics such as the number of samples, as well as the number of rules that the analyst intends to generate. For example, the authors in [12] consider minimum support values ranging from 2% to 0.5% for a dataset that contains 100,000 samples. In fact, increasing the value for minimum support will boost the statistical significance of the rules; however, it will lead to fewer rules to be generated.

6.2 *Minimum confidence*: For the purpose of determining the strength of rules, minimum value for confidence ought to be defined. Similar to minimum support value, increasing minimum confidence value results in finding stronger but fewer rules. Again, the selection of this value is at analyst's discretion.

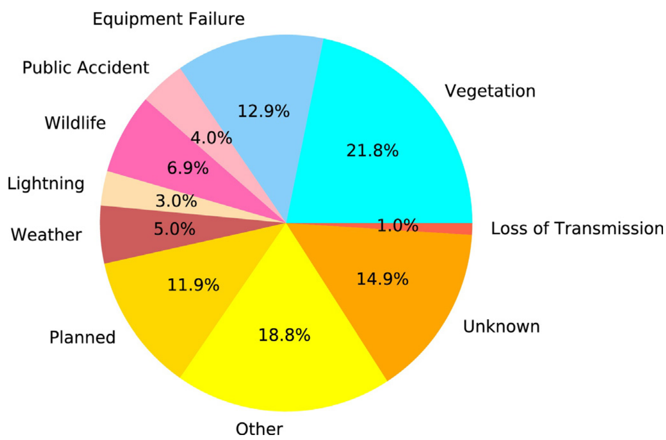

Fig. 4. Distribution of different faults for Charlotte between the years 2009 and 2014.

6.3 *Setting RHS of rules*: By applying the Apriori algorithm, numerous rules might be found; however, since the purpose of the study is to find the patterns for faults' causes, those rules are desired whose RHS is the target fault cause. Therefore, for each sub-dataset, the RHS of the rules is to be set to dummy variable that is created for the fault cause. For instance, the RHS of the rules for analyzing vegetation sub-dataset is to be set to VEGETATION1.

(7) *Inspecting rules*: As mentioned, applying the Apriori algorithm results in generating a variety of rules. In order to identify best rules, which show the patterns for fault cause, they are required to be sorted by the lift value. Afterward, those with the highest lift value are selected and interpreted by the analyst.

5. Case study

In order to demonstrate how the proposed approach is to be properly implemented in practice, a case study is investigated in this section. The main goal is to characterize vegetation, animal, equipment failure, lightning and, public accident-related faults that commonly occur in and around Charlotte region according to their underlying causes and to identify significant variables that strongly impact the frequency of these faults. To carry out this task, the proposed procedure is followed step-by-step, and results are achieved and discussed. It is worth mentioning that various factors could affect the frequency of different faults, and based on the results, it is clear that those factors could have high dependency on the geographical location of the distribution system, the environmental characteristics of the region, etc. Therefore, it would not always be possible to generalize the rules that we achieve in this paper. However, we claim that the proposed approach may be applied on any fault dataset to find the patterns of interest.

5.1. Implementation

The proposed algorithm is implemented as follows.

(1) *Data collection and preparation*: In this paper, fault data collected by Duke Energy is utilized. The data includes information on faults that occurred in Charlotte, North Carolina region, between the years 2009 and 2014. The data is comprised of almost 55,000 samples and almost 20 features. The faults occurred due to various causes. Fig. 4 demonstrates the distribution of different faults based on their causes.

1.1 *Obtaining data for various features*: Every time a fault occurs in distribution systems within the domain of Duke Energy, several pieces of information related to that distribution fault are recorded into a dataset as one record entry. Each fault record is described with almost 20 features. Nevertheless, among the features, some are irrelevant to the purpose of this study; hence, they are excluded from this analysis. Those features that are selected to be included are: weather condition, time, voltage level, circuit number, faulty phases, and activated clearing device. The time is split by the authors into three groups of the month, time of day, and weekday. Furthermore, the authors have added four additional useful features of temperature, humidity percentage, dew, and wind speed to the available data from external sources.

1.2 *Data cleansing*: The data set contains a few number of input errors and duplicate data. Therefore, by exploring data set in the R programming language, all errors are identified and removed.

1.3 *Dealing with missing values*: There is a large number of missing values in the dataset. In order to effectively deal with them, the "randomForestSRC" algorithm developed by [18] in the *R* programming language is implemented in this paper. In this method, the missing data is imputed by randomly drawing values from non-missing values. For this purpose, a random forest is grown and is used to impute missing data.

(2) *Converting continuous features to categorical factors*: In what follows, all features are fully described, and it is explained how they are converted to categorical factors. It is worth mentioning that the exact name of the feature, which is utilized in different steps of the algorithm is provided in the parenthesis.

- *Weather condition (WEATHER)*: Each fault is associated with a weather condition. Weather conditions are categorized into 10 groups: calm, extreme cold, rain, wind, the combination of wind, rain, and lightning, lightning, snow, ice, extreme heat, and the combination of wind and rain. These are represented by factors 0–9, respectively.
- *Month (MONTH)*: Month can affect the frequency of faults with different causes. Months are represented by numbers 1–12.
- *Weekday (WEEKDAY)*: Weekday can make impact on some specific faults such as public accident-related faults as the car traffic differs between days. In addition, the authors demonstrated in [14] that weekday takes high importance with regard to identifying equipment failures. Therefore, this feature is included in the analysis, which the days are represented by numbers 1–7, where Monday is represented by 1.
- *Time of day (TIME)*: Time of day is highly useful for fault root cause analysis since several environmental parameters are associated with it. In order to add this feature to each fault sample, first, different time intervals are created. 6 am–12 pm, 12 pm–5 pm, 5 pm–8 pm, 8 pm–12 am, 12 am–6 am are defined as morning, afternoon, evening, night and midnight, respectively. Then, the aforesaid groups are represented by 1–5, respectively.
- *Clearing device activated (CD)*: When a fault occurs, the information about the device that was activated to clear it is recorded as a feature. This feature is extremely valuable as it helps in finding the patterns for post-fault situations. There are 12 clearing devices in the dataset: substation device, feeder breaker, line recloser, line fuse, transformer fuse, transformer CSP, HPP/MHO breaker, service/secondary device, disconnect device, jumper, sectionlizer, and transmission device. These devices are represented by 0–11, respectively.
- *Voltage level (VOLTAGE)*: Distribution systems are comprised of different voltage levels, which all level are exposed to various faults. There are five different voltage levels for existing distribution circuits in Charlotte: 2 kV, 4 kV, 7 kV, 12 kV, and 24 kV. In the dataset, these groups are represented by numbers 1–5, respectively.
- *Faulty phases (PHASE)*: Different line phases can be affected by a fault. Duke Energy captures the faulty phases and records them in the dataset. Each possible combinations of faulty phases are represented by a number. All combinations are included in the analysis; however, two of them, namely three-phase and one-

phase are observed to be necessary for interpreting the results of this study. These two combinations are represented by number 5 and 10, respectively.

- *Circuit (CIRCUIT)*: There are more than 160 distribution circuits associated with fault samples available in Duke Energy dataset. These circuits differ in terms of age, length, location and voltage level. Each circuit is represented by a specific number.

Different environmental features including wind speed, humidity, dew point value, and temperature can exert significant effect on fault frequency. Therefore, they are included in the analyses, and are described as follows.

- *Wind speed (WIND)*: Faults occur in a broad range of wind speed. In this dataset, the range covers wind speeds of 1–19 m/s. The wind speed is categorized into 7 groups of [1,3.5), [3.5,6), [6,8.5), [8.5,11), [11,13.5), [13.5,16), and [16,19] and then represented by categorical factors of 1–7, respectively.
- *Humidity (HUMIDITY)*: The range for humidity varies from 20% to 96%. Similar to wind, humidity values are categorized in 7 groups. The categories are [20,36), [36,46), [46,56), [56,66), [66,76), [76,86), and [86,96]. These groups are represented by categorical factors of 1–7, respectively.
- *Dew (DEW)*: Dew values associated with the faults in the dataset range between 0 °F and 74 °F, which are categorized into 7 groups of [0,13), [13,23), [23,33), [33,43), [43,53), [53,63), and [63,73] and represented by 1–7, respectively.
- *Temperature (TEMPERATURE)*: Similar to the other environmental variables discussed, 7 categories are created for temperature. Temperature values range through 16 °F and 90 °F. The created categories are [16,30), [30,40), [40,50), [50,60), [60,70), [70,80), and [80,90] and represented by 1–7, respectively.

(3) *Defining new dataset for each fault cause*: The next step is to create a new dataset for each target fault cause. As mentioned earlier, in this case study, vegetation, animal (wildlife), equipment failure, lightning and public accident-related faults will be explored; consequently, five sub-datasets will be created. In order to carry this out, the procedure illustrated in Fig. 3 is followed five times.

(4) *Generating synthetic data for each sub-dataset*: The next step is to generate synthetic data for each sub-dataset. In order to conduct this task, SMOTE algorithm which was explained earlier is utilized. To perform this algorithm, Random Over-Sampling Examples (ROSE) package developed by [19] in the *R* programming language is implemented five times, one time for each sub-dataset. After the method is applied, the percentage of instances in each sub-dataset whose fault cause value is represented by 1 increases. In general, the decision on the amount of synthetic data to be generated depends on the data size, types of data, features, as well as other factors. Therefore, different amounts of synthetic data should be generated and tested to understand how it affects the performance of the data analytics algorithm. For example, the authors in [15] consider different amounts of synthetic data ranging between 100% and 500%. In this paper, the number of synthetic data varied between 200% and 700% based on the fault type. By creating such synthetic data, each sub-dataset would contain sufficient amounts of information relating to each fault cause.

(5) *Converting categorical factors to dummy variables*: Then, by following the procedure explained in Section 4, part 4, all categorical factors are converted to dummy variables for all sub-datasets. For example, as mentioned earlier, one feature is month, which takes values of 1–12. These values are converted to MONTH1, . . ., MONTH12. This should be performed for all available features. Similarly, the same procedure ought to be followed for the column that demonstrates the fault cause.

(6) *Mining association rule*: Generated sub-datasets, which look like basket data type transactions, are ready to be investigated by association rule mining. In order to mine these sub-datastes, "arules" package [20], which is based on Apriori algorithm is implemented in *R* programming language. Necessary constraints are specified as follows.

6.1 *Minimum support*: The minimum support to generate large itemsets in this study is selected to be 0.0018, which is equivalent to 100 occasions. We believe that if an itemset appears more than 100 times, it can be assumed to be frequent. In fact, it is decided not to select a high value for the minimum support as we prefer that the algorithm generates more rules. These rules could be further investigated and interpreted to select the most important ones. It is worth mentioning that selecting a low value is always a better option compared to selecting higher minimum support because the latter option could result in missing possible important rules.

6.2 *Minimum confidence*: The minimum confidence is selected to be 70%. As mentioned earlier, the selection of this value depends on analyst's decision.

6.3 *Setting RHS of rules*: For each sub-dataset, the RHS of the rules is set to dummy variable that is created for that fault cause. For instance, the RHS of the rules for analyzing vegetation sub-dataset is set to VEGETATION1.

(7) *Inspecting rules*: In order to find the top-ranked rules, which better show the patterns for fault causes, all resultant rules are inspect by applying the inspect function defined in the Apriori package. The procedure is to sort all rules and identify the high-quality rules whose lift values is significant. In this paper, those rules whose lift is greater than 1.4 are selected and presented.

5.2. Results and discussion

In what follows, the results of association rule mining for vegetation, animal, equipment failure, public accident, and lightning-related faults are presented and discussed. It is worth mentioning that the LHS, RHS, support, confidence and lift values of rules are provided. Also, the rules are sorted based on their lift value.

(1) *Vegetation-related faults*: The strongest rules achieved for vegetation-related faults are provided in Table 1. It is worth mentioning that there are several rules that include location (i.e. circuit number) in their LHS; however, due to the security reasons they are not provided in the table. According to the results, it can be understood that extreme heat, ice, snow, and combinations of wind and rain are the most frequent weather conditions for vegetation-related faults. Moreover, very low temperature, very high humidity, high wind, and location play crucial roles in such faults. Also, line reclosers and sectionlizers are the primary devices that clear these faults.

(2) *Animal-related faults*: Table 2 provides the considerable rules extracted for animal-related faults. Similar to vegetation-related rules, the animal-related rules that contain location in their LHS are not provided in the table. However, based on the analysis, it is noticed that the location make a great impact on these faults. In summary, based on Table 2, it can be concluded that animal-related faults are highly dependent on month. In fact, most of such faults in Charlotte, North Carolina region, occur during the last three months of the year. Furthermore, it can be understood that dew points that are above average make a significant contribution to such faults. Finally, it can be observed that most of the animal-related faults are cleared by transformer fuse.

(3) *Equipment failure-related faults*: The rules achieved for equipment failures are provided in Table 3. Based on this table, it is noticed that extreme cold and extreme hot weather conditions

Table 1
Results of association rule mining for vegetation-related faults.

LHS		RHS	Support	Confidence	Lift
{WEATHER8, WIND4}	⇒	{VEGETATION1}	0.003337871	0.9728261	1.944528
{WEATHER8, CD3, DEW4}	⇒	{VEGETATION1}	0.002480094	0.9500000	1.898902
{WEATHER8, HUMIDITY5}	⇒	{VEGETATION1}	0.003636228	0.9466019	1.892110
{MONTH2, CD2, TEMPERATURE2}	⇒	{VEGETATION1}	0.001957969	0.9459459	1.890799
{WEATHER17, TEMPERATURE6, WIND4}	⇒	{VEGETATION1}	0.002237679	0.9302326	1.859390
{WEATHER17, HUMIDITY6, WIND4}	⇒	{VEGETATION1}	0.002424152	0.9154930	1.829928
{TIME5, HUMIDITY7, DEW7}	⇒	{VEGETATION1}	0.002405505	0.9084507	1.815852
{WEATHER7, TEMPERATURE2, HUMIDITY6}	⇒	{VEGETATION1}	0.002890335	0.9011628	1.801284
{WEATHER17, WIND7}	⇒	{VEGETATION1}	0.002013911	0.8503937	1.699805
{MONTH2, CD2}	⇒	{VEGETATION1}	0.002386857	0.8951049	1.789176
{CD2, HUMIDITY7}	⇒	{VEGETATION1}	0.002591978	0.8424242	1.683875
{CD10}	⇒	{VEGETATION1}	0.003804054	0.7669173	1.532948
{WEATHER7}	⇒	{VEGETATION1}	0.009006657	0.7606299	1.520381
{WIND7}	⇒	{VEGETATION1}	0.009808492	0.7377279	1.474603

Table 2
Results of association rule mining for animal-related faults.

LHS		RHS	Support	Confidence	Lift
{MONTH6, CD4, DEW5}	⇒	{ANIMAL1}	0.002647920	0.9594595	1.917810
{MONTH9, CD4, DEW5}	⇒	{ANIMAL1}	0.005183956	0.9553265	1.909549
{MONTH12, CD4, HUMIDITY5}	⇒	{ANIMAL1}	0.008316706	0.9550321	1.908961
{MONTH10, TIME3, CD4}	⇒	{ANIMAL1}	0.005370429	0.9442623	1.887434
{MONTH12, CD4, DEW5}	⇒	{ANIMAL1}	0.004568594	0.9423077	1.883527
{MONTH12, CD4, TEMPERATURE5}	⇒	{ANIMAL1}	0.002088500	0.9411765	1.881265
{MONTH11, PHASE10, CD4}	⇒	{ANIMAL1}	0.011244336	0.9363354	1.871589
{MONTH10, CD4, DEW4}	⇒	{ANIMAL1}	0.007272456	0.9285714	1.856070
{MONTH11, CD4, WIND2}	⇒	{ANIMAL1}	0.013239599	0.9281046	1.855137
{MONTH11, CD4}	⇒	{ANIMAL1}	0.034385664	0.8995122	1.797985
{MONTH12, CD4}	⇒	{ANIMAL1}	0.029164413	0.8901537	1.779279
{MONTH10, CD4}	⇒	{ANIMAL1}	0.025062002	0.8604353	1.719876
{MONTH11}	⇒	{ANIMAL1}	0.065452104	0.7164727	1.432118

Table 3
Results of association rule mining for equipment failure-related faults.

LHS		RHS	Support	Confidence	Lift
{WEATHER1, WEEKDAY2, WIND2}	⇒	{EQUIPMENT1}	0.004307532	0.9130435	1.825032
{WEATHER1, MONTH1, WEEKDAY2}	⇒	{EQUIPMENT1}	0.004177000	0.9105691	1.820086
{VOLTAGE24, WEEKDAY7, CD7}	⇒	{EQUIPMENT1}	0.005780670	0.9037901	1.806536
{TEMPERATURE 7, HUMIDITY3, WIND2}	⇒	{EQUIPMENT1}	0.002797098	0.8928571	1.784683
{WEATHER1, PHASE10, HUMIDITY2}	⇒	{EQUIPMENT1}	0.001920674	0.8879310	1.774836
{WEATHER1, PHASE10, DEW1}	⇒	{EQUIPMENT1}	0.002517389	0.8766234	1.752234
{WEATHER1, TEMPERATURE1, DEW1}	⇒	{EQUIPMENT1}	0.007011394	0.8764569	1.751901
{WEATHER1, TEMPERATURE1}	⇒	{EQUIPMENT1}	0.007011394	0.8744186	1.747827
{WEATHER1}	⇒	{EQUIPMENT1}	0.007570813	0.8565401	1.712090
{TEMPERATURE1}	⇒	{EQUIPMENT1}	0.016018051	0.7182274	1.435625
{DEW1}	⇒	{EQUIPMENT1}	0.023141328	0.7095483	1.418277

Table 4
Results of association rule mining for public accident-related faults.

LHS		RHS	Support	Confidence	Lift
{VOLTAGE24, CD9}	⇒	{ACCIDENT1}	0.002219031	0.9370079	1.872933
{MONTH10, CD1}	⇒	{ACCIDENT1}	0.003132750	0.9230769	1.845087
{MONTH11, CD1}	⇒	{ACCIDENT1}	0.002983572	0.9142857	1.827515
{MONTH10, CD2}	⇒	{ACCIDENT1}	0.003170045	0.8947368	1.788440
{MONTH2, WIND7}	⇒	{ACCIDENT1}	0.003244634	0.8923077	1.783584
{HUMIDITY3, WIND7}	⇒	{ACCIDENT1}	0.003244634	0.8923077	1.783584
{WEEKDAY5, WIND7}	⇒	{ACCIDENT1}	0.003244634	0.8923077	1.783584
{WEATHER2, DEW2}	⇒	{ACCIDENT1}	0.002107148	0.8897638	1.778499
{PHASE5, CD9}	⇒	{ACCIDENT1}	0.009025304	0.8897059	1.778384
{CD2, TEMPERATURE3}	⇒	{ACCIDENT1}	0.005911201	0.8781163	1.755218
{CD1, DEW5}	⇒	{ACCIDENT1}	0.006489268	0.8721805	1.743353
{MONTH1, CD1}	⇒	{ACCIDENT1}	0.003710817	0.8689956	1.736987
{TIME5, CD9}	⇒	{ACCIDENT1}	0.003020866	0.8617021	1.722409
{CD1}	⇒	{ACCIDENT1}	0.032502284	0.7678414	1.534796
{CD9}	⇒	{ACCIDENT1}	0.012363175	0.7542662	1.507661

Table 5
Results of association rule mining for lightning-related faults.

LHS		RHS	Support	Confidence	Lift
{WEATHER5, MONTH3}	⇒	{LIGHTNING1}	0.002871688	0.9685535	1.935988
{WEATHER5, CD4}	⇒	{LIGHTNING1}	0.018442203	0.9509615	1.900824
{WEATHER5, HUMIDITY4}	⇒	{LIGHTNING1}	0.010852742	0.9494290	1.897761
{WEATHER5, TIME2}	⇒	{LIGHTNING1}	0.012288586	0.9374111	1.873739
{WEATHER5, DEW6}	⇒	{LIGHTNING1}	0.005445018	0.9240506	1.847034
{WEATHER5, DEW7}	⇒	{LIGHTNING1}	0.010144144	0.9189189	1.836776
{WEATHER5, HUMIDITY6}	⇒	{LIGHTNING1}	0.018927033	0.9168925	1.832726
{WEATHER5, CD3}	⇒	{LIGHTNING1}	0.003263282	0.9114583	1.821864
{WEATHER5, VOLTAGE24}	⇒	{LIGHTNING1}	0.029444123	0.9090386	1.817027
{WEATHER5, MONTH7}	⇒	{LIGHTNING1}	0.158539542	0.7100977	1.419375

have a huge impact on equipment failures. In fact, a majority of these faults occur when the temperature, humidity, and dew point values are either very low or very high. Furthermore, it is found that circuits with the highest voltage (i.e. 24 kV) are main locations for equipment failures. Also, weekday makes a contribution to these faults.

(4) *Public accident-related faults*: Table 4 summarizes the rules with the highest strength for faults with regards to public accidents. According to the analysis, these faults are highly affected by location (circuit number); however, as mentioned, due to security reasons, the rules that have circuit number are not presented in the table. Based on the rest of results, it is understood that rain weather condition, high winds, low visibility (i.e. TIME5), and month have great impacts on these type of faults. Also, feeder breaker and jumpers can be identified as main devices that clear accident-related faults.

(5) *Lightning-related faults*: Table 5 demonstrates the important rules for lightning-related faults. It is obvious that lightning situation is a necessity for these type of faults. However, based on the table, it is understood that month, high voltage, high humidity and high dew points also can affect lightning faults. Moreover, location plays a key role for these faults; nevertheless, due to security reasons, circuit numbers are not presented in the table. With respect to clearing device, it can be mentioned that the majority of faults due to the lightning are cleared by line fuse and line recloser.

6. Conclusions

In this paper, a novel approach for fault root cause analysis by utilizing association rule mining was proposed. For this purpose, a detailed procedure, which includes several steps was provided to effectively deal with required data preparation, practical problems associated with real-world fault data set, and implementation of association rule mining. In order to show the application of the proposed approach in practice, a case study was selected and investigated. By employing the proposed approach various faults including vegetation, animal, equipment failure, lightning, and public accident-related faults were characterized according to their underlying causes, and variables that strongly impact their frequency were identified. Moreover, the co-occurrence of these variables was investigated. The results demonstrate that applying this approach successfully yields revealing insights into the causal structures and frequent patterns of different faults that commonly occur in power distribution systems.

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